

Homework 4

Name:

Problem 1

Trapezoidal Integration

Given the following two functions of x

$$f_1(x) = x + 2 \quad f_2(x) = x^3 + 1$$

- (a) Compute the integral of f_1 from 0 to 2 by hand:

$$\int_0^2 f_1(x) \, dx$$

- (b) Compute the integral of f_2 over the same interval by hand:

$$\int_0^2 f_2(x) \, dx$$

- (c) Write a Python function to implement the trapezoidal rule for integration and use it to compute the two integrals above. (Your function should take four numeric inputs: a , b , $f(a)$, and $f(b)$.)
- (d) Do your trapezoidal rule results match your previous hand computations? Why or why not?

Problem 2**Integrating Radiation Dose**

Portable radiation measurement tools called electronic personal dosimeters (EPDs) are used in nuclear power facilities to measure the total dose of radioactive material each individual experiences. Suppose a worker turns in a dosimeter at the end of the shift which contains the following sensor data:

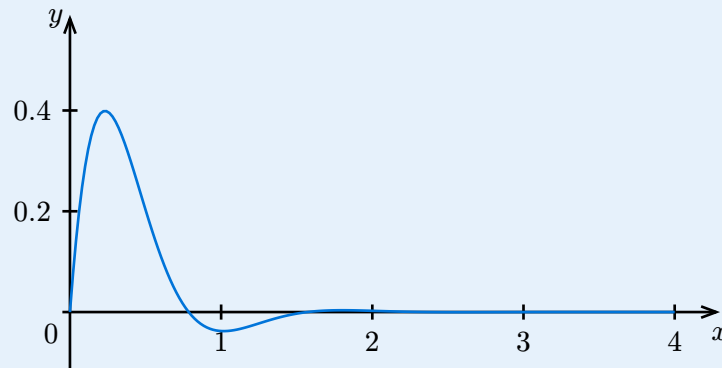
Time (hrs)	Radiation Dose Rate ($\mu\text{Sv/hr}$)
0	2.015
1	3.822
2	4.002
3	3.989
4	1.881
5	15.033
6	61.890
7	17.402

- (a) Use your choice of integration method to integrate the data. This gives you the total dose (in microSieverts) experienced by this radiation worker during this shift.
- (b) The federal limit for radiation workers in the US is 50,000 microSieverts per year. Suppose this worker experiences the same radiation dose you computed above every day for a full year. Would his or her dose exceed the federal limit?
- (c) What is the truncation error term associated with your selected integration method? Explain why you would or would not trust this result.

Problem 3

Comparing Integration Methods

The motion of a particular spring-damper system along its primary axis of motion is described by the equation $y(x) = e^{-3x} \sin(4x)$.



The area under this curve is almost exactly $\frac{4}{25}$. This number describes the total deviation from equilibrium caused an external disturbance.

- Write a Python function to implement the composite trapezoidal rule with N subintervals. Use at least 10 subintervals to approximate the total area under the curve on the interval $(0, 3.7)$.
- Write a Python function to implement composite 4-point Gaussian quadrature with N subintervals. Use 3 subintervals to approximate the total area under the curve on the interval $(0, 3.7)$.
- Write a Python function to implement adaptive quadrature. Use it to compute the result to within 10^{-6} .
- Answer the following questions about your results:
 - How close are your results to the true value of $\frac{4}{25}$?
 - Which method gives the best result?
 - Does your answer to the previous question line up with what you expected? Why or why not?

Problem 4**5-Point Differentiation**

Let $f(x) = 8x^8 - 12x^7 - 62x^6 + 89x^5 + 156x^4 - 215x^3 - 126x^2 + 186x$.

- (a) Compute the derivative $f'(x)$ analytically (by hand).
- (b) Write a Python function to compute the derivative using the 5-point midpoint method:

$$f'(x) \approx \frac{1}{12h} [f(x-2h) - 8f(x-h) + 8f(x+h) - f(x+2h)]$$

Your Python function should take a callable f and float h as inputs and return a callable function of x .

- (c) Plot the analytical derivative against your computed derivative with $h = \frac{1}{2}$, and $h = \frac{1}{16}$. Use a legend to indicate which line is which.

Problem 5**Rocket Peak Acceleration**

The position of a rocket (in kilometers) during the liftoff stage can be approximated by the function

$$y(t) = -156 - \frac{1}{8}e^{t-447} + 78\left(e^{-\frac{t}{500}} + e^{\frac{t}{500}}\right)$$

- (a) Use a numerical method of your choice to construct a function for the second derivative $y''(t)$. This describes the acceleration of the rocket. Plot the position and acceleration from 0 to 450 seconds.
- (b) Use your second derivative with an optimization method to find the maximum acceleration that rocket experiences before the first stage shuts off at $t = 450$.